

Dhruv Varshney | 23EX10013

Bachelor of Science in Exploration Geophysics (4-Year), Dept. of Geology and Geophysics
Pursuing Minor in Computer Science and Engineering
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Education

Year	Degree/Exam	Institution	CGPA/Marks
2027	4 Year B.S	IIT Kharagpur	8.3*/10
2023	Central Board of Secondary Education (CBSE)	Kendriya Vidyalaya IIT Kharagpur	91.2%
2021	Central Board of Secondary Education (CBSE)	Kendriya Vidyalaya IIT Kharagpur	97.6%

* CGPA till 5th Semester

Interests

- **Applied Machine Learning:** Large Language Models, speech recognition, and representation learning.
- **Data Engineering:** Scalable ETL pipelines, data quality automation, and analytics infrastructure.
- **Statistical Modeling:** Experiment design, probabilistic analysis, and decision science.
- **Information Retrieval:** Semantic search, knowledge graph enrichment, and entity resolution.
- **Storytelling with Data:** Dashboarding, insight communication, and user-centric analytics.

Experience

Analyst Programmer - Xceedance Consulting Private Limited

May '25 - Oct '25

Extraction of information from insurance data using **LLM** workflows and domain-specific normalization.

- Generated diverse synthetic datasets with **genAI** to cover edge-case policy structures.
- Performed **exploratory data analysis** and data cleaning, improving downstream data reliability.
- **Normalized** entity names and attributes to enforce consistency across the claims knowledge base.
- Built and evaluated a **few-shot model** on the **Azure AI API** to classify critical insurance entities.

Vedvani: Transcribing Spoken Sanskrit Shlokas

May '24 - Nov '24

Under Prof. Pawan Goyal, IIT Kharagpur

Engineered a **speech-to-text pipeline** by scraping Sanskrit web text with **svaras** and training a custom **Transformer model**.

- Scraped and aligned transcripts with audio recordings using **BeautifulSoup** and bespoke preprocessing scripts.
- Cleaned and normalized text corpora to preserve **svaras** and improve phoneme level accuracy.
- Fine-tuned a **Wav2Vec2 model** for low-resource Sanskrit, introducing data augmentation for robustness.
- Measured quality using **Word Error Rate (WER)** and **Character Error Rate (CER)**, achieving consistent gains.

Projects

Real-Time AI Emergency Evacuation System | Self Project

Project

Engineered a computer vision system that dynamically calculates optimal evacuation routes in real-time from live video feeds, mapping detections to a digital floor plan.

- Integrated **YOLOv11** for human detection and **HSV** color-thresholding for hazard simulation, processing frames at **30 FPS**.
- Applied **homography** (perspective transformation) to map 3D camera coordinates onto a 2D digital floor plan with high spatial accuracy.
- Implemented the **A* (A-Star) pathfinding** algorithm to dynamically regenerate safe routes in under **50 ms** when moving obstacles or hazards are detected.
- Optimized real-time performance using **OpenCV** and **NumPy** for low-latency processing and visualization overlays.
- Designed a modular pipeline for sensor integration, floorplan overlays, and alerting to support rapid evacuation decision-making.

Prediction of House Prices | Kaggle

Competition

Developed and optimized regression models on Kaggle's advanced house prices dataset, achieving competitive leaderboard placement. [View Competition](#)

- Engineered **200+ features** from 79 original variables using statistical imputation for missing values (28% in some features) and one-hot encoding for 43 categorical variables, improving model performance by 30%.
- Developed and compared 6 regression models (Linear Regression, Ridge, Lasso, Random Forest, Gradient Boosting, XGBoost), achieving **90% R² score** and **\$25,450 RMSE**.
- Implemented **Bayesian Optimization** with scikit-optimize to tune 9 XGBoost hyperparameters across 30 iterations with 5-fold cross-validation, reducing prediction error by 50%.
- Conducted comprehensive exploratory data analysis with 15+ visualizations (correlation heatmaps, distribution plots) and feature importance analysis, identifying top 20 price-driving factors.
- Achieved **Kaggle leaderboard rank: 2,111 out of thousands** in the House Prices: Advanced Regression Techniques competition.

Spaceship Titanic | Kaggle

Competition

Built an end-to-end classification workflow for Kaggle's Spaceship Titanic challenge. [View Competition](#)

- Engineered a full pipeline covering feature creation (amenity aggregations, cabin parsing), missing-value strategy, and aligned one-hot encoding for submission data.
- Benchmarked and tuned **Random Forest, SVM, and Naive Bayes** classifiers with **GridSearchCV**, generating calibrated probabilities and ROC/confusion-matrix diagnostics.
- Achieved **Kaggle rank 1,582** with public leaderboard score **0.75403**, producing automated submission artifacts for repeatable uploads.

Titanic Survival Prediction | Kaggle

Competition

Developed an optimized XGBoost model for predicting Titanic passenger survival, achieving strong performance through comprehensive EDA and feature engineering. [View Competition](#)

- Analyzed 891 training samples across 12 features, uncovering survival patterns: females (74% survival) vs. males (19%), and 1st class (63%) vs. 3rd class (24%), informing strategic feature selection.

- Built production-ready preprocessing pipeline with statistical imputation for 20% missing Age data, one-hot encoding for 7 categorical variables, and robust train-test feature alignment.
- Developed **XGBoost classifier** with 500 estimators and 6 tuned hyperparameters (learning rate 0.05, max depth 4, L2 regularization), achieving **ROC-AUC 0.82-0.88** and identifying Sex, Fare, and Age as top predictive features.
- Achieved **Kaggle rank 3,916 with accuracy 0.77511**, generating 418 test predictions with comprehensive evaluation suite including confusion matrix, ROC curve, and feature importance analysis.

Awards and Achievements

- Ranked among the **top 6.4% in JEE Advanced 2023** and **top 1.9% in JEE Mains 2023**.
- Ranked among the **top 1% in WBJEE** (West Bengal Joint Entrance Exam) among more than 1 Lakh students.
- Achieved a **peak rating of 1400+ (Specialist) on Codeforces** demonstrating strong problem-solving skills.
- **Solved 150+ Leetcode Problems** consistently improving coding abilities and logical reasoning.
- Secured **4th place overall** in the **Open IIT Data Analytics Competition 2024**.

Skills and Expertise

- **Languages:** Python, SQL, C++, Bash
- **ML & AI:** scikit-learn, TensorFlow, Hugging Face Transformers, OpenCV, spaCy
- **Data Ops:** Feature engineering, data validation, experiment tracking, ML evaluation
- **Visualization:** Matplotlib, Seaborn, Plotly, dashboard storytelling

Coursework Information

- **Computer Science and Mathematics:** Programming and Data Structures (Lab and Theory), Linear Algebra, Advanced Calculus, Partial Differential Equation, Probability and Statistics, Transform Calculus
- **MOOCs:** **Machine Learning Specialization** by Andrew Ng (Coursera), Audio Course by Hugging Face, Matlab Onramp by Mathworks, Strivers A2Z Course

Positions of Responsibility

- **Head of Design and Content Writing | Click Kgp:** Proficient in multiple photography styles including event, lifestyle, and wildlife, with advanced post-production skills.